

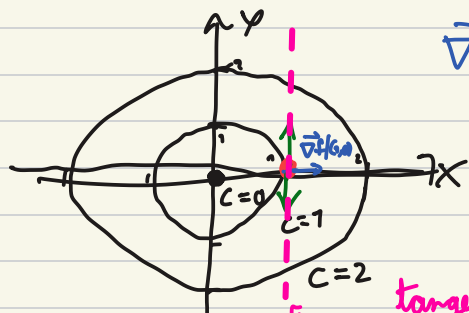
March 2, 2026

Continuing the example at the end of last lecture:

e.g. $f(x,y) = \sqrt{x^2 + y^2} = c$

$$\vec{\nabla} f = \left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}} \right)$$

$$\vec{\nabla} f|_{(1,0)} = (1, 0)$$



On what directions $\hat{v}$
 $\vec{0} = \vec{\nabla} f|_{(1,0)} \cdot \hat{v} = 0$? $(0,1), (0,-1)$

tangent line $x=1$ or $(1,0) \cdot (x-1, y) = 0$

How do we interpret this?

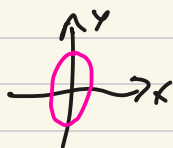
If we approximate the level curve $f(x,y) = c$ (for $c=1$) at the point $\vec{a} = (x_0, y_0) = (1, 0)$ by its tangent line, then we stay on the same level curve (at least in its linear approximation, i.e. the tangent line) if we move in a direction perpendicular to $\vec{\nabla} f|_{\vec{a}}$.

Thm: Suppose $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ (with X open) is differentiable at $\vec{a}$ and has $\vec{\nabla} f|_{\vec{a}} \neq \vec{0}$. Then the equation $0 = \vec{\nabla} f|_{\vec{a}} \cdot (\vec{x} - \vec{a})$

defines the tangent hyperplane at $\vec{a}$ to the level set given by $f(\vec{x}) = c$ (for $c = f(\vec{a})$)

e.g. tangent line to ellipse $\frac{x^2}{2} + \frac{y^2}{4} = 1$ at $(1, \sqrt{2})$
 $f(x,y) = \frac{x^2}{2} + \frac{y^2}{4}$ $\vec{\nabla} f = (x, \frac{y}{2})$ $\vec{\nabla} f|_{(1,\sqrt{2})} = (1, \frac{\sqrt{2}}{2})$

tangent line $0 = (1, \frac{\sqrt{2}}{2}) \cdot (x-1, y-\sqrt{2}) = x-1 + \frac{\sqrt{2}}{2} y - 1 = 0$



Applications of Chain Rule

8. The intensity I of a source of light, such as a light bulb, as perceived by someone standing some distance away from it is given by the formula

$$I = \frac{P}{4\pi R^2},$$

where P is the power of the source measured in watts and R the distance to the source measured in meters. (Intensity has units of watts per square meter.) Suppose you begin to walk away from the source, and that after 10 seconds the power is increasing at a rate of 0.1 watts per second, and that at the same time you are moving away at a rate of 1.4 meters per second. Suppose also that at this instant the power of the source is 100 watts and you are 10 meters away from it.

- At what rate is the intensity of the light changing?
- Suppose the power was instead increasing at a rate of 15 watts per second. How quickly would you have to move away from the source in order to have the intensity change at the same rate as in part (a)?

$$a) I(P, R) : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}$$

$$P(t) : \mathbb{R} \rightarrow \mathbb{R}$$

$$R(t) : \mathbb{R} \rightarrow \mathbb{R}$$

$$\frac{\partial I}{\partial t} = \frac{df}{dt} = Df$$

$$f(t) = I \circ (P(t), R(t))$$

$$= DI|_{(P,R)} \cdot D(P,R)|_t$$

$$= \left(\frac{\partial I}{\partial P} \quad \frac{\partial I}{\partial R} \right) \begin{pmatrix} \frac{\partial P}{\partial t} \\ \frac{\partial R}{\partial t} \end{pmatrix} = \frac{\partial I}{\partial P} \frac{\partial P}{\partial t} + \frac{\partial I}{\partial R} \frac{\partial R}{\partial t} = \frac{\partial I}{\partial t}$$

$$\frac{\partial P}{\partial t} = 0.1 \text{ W/s}$$

$$P = 100$$

$$\frac{\partial R}{\partial t} = 1.4$$

$$R = 10$$

$$\frac{\partial I}{\partial P} = \frac{1}{4\pi R^2}$$

$$\frac{\partial I}{\partial R} = \frac{-2P}{4\pi R^3}$$

$$\frac{\partial I}{\partial t} = \frac{1}{4\pi (10)^2} \cdot 0.1 + \frac{-2(100)}{4\pi (10)^3} \cdot 1.4 = \frac{1}{4\pi 10^3} + \frac{-280}{4\pi 10^3} = \frac{-279}{4\pi 10^3}$$

14. Suppose that $z = f(x + y, x - y)$ has continuous partial derivatives with respect to $u = x + y$ and $v = x - y$. Show that

$$\frac{\partial z}{\partial x} \frac{\partial z}{\partial y} = \left(\frac{\partial z}{\partial u} \right)^2 - \left(\frac{\partial z}{\partial v} \right)^2.$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} (1) + \frac{\partial z}{\partial v} (1)$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = \frac{\partial z}{\partial u} (1) + \frac{\partial z}{\partial v} (-1)$$

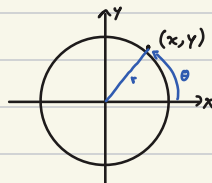
$$\left(\frac{\partial z}{\partial x} \right) \left(\frac{\partial z}{\partial y} \right) = \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) = \left(\frac{\partial z}{\partial u} \right)^2 - \left(\frac{\partial z}{\partial v} \right)^2$$

Optional topic (won't be tested)

Differential operator $\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$ in polar coordinates

In $\mathbb{R}^2$, polar coordinates are given by:

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$



Suppose w is a quantity that depends on (x, y) by $w = f(x, y)$ (for some differentiable function f)

Then w depends on (r, θ) by $w = g(r, \theta) = f(x(r, \theta), y(r, \theta))$ (g is also differentiable by the chain rule)

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r}$$

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial \theta}$$

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \cos \theta + \frac{\partial w}{\partial y} (\sin \theta)$$

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x} (-r \sin \theta) + \frac{\partial w}{\partial y} (r \cos \theta)$$

$$\frac{\partial x}{\partial r} = \cos \theta \quad \frac{\partial x}{\partial \theta} = -r \sin \theta$$

$$\frac{\partial y}{\partial r} = \sin \theta \quad \frac{\partial y}{\partial \theta} = r \cos \theta$$

Addendum: Now we solve for $\frac{\partial w}{\partial x}$ and $\frac{\partial w}{\partial y}$ in terms of r, θ only:

$$r^2 \cos \theta \frac{\partial w}{\partial r} - r \sin \theta \frac{\partial w}{\partial \theta} = r^2 \cos^2 \theta \frac{\partial w}{\partial x} + r \sin^2 \theta \frac{\partial w}{\partial x} = r^2 \frac{\partial w}{\partial x}$$

$$r^2 \sin \theta \frac{\partial w}{\partial r} + r \cos \theta \frac{\partial w}{\partial \theta} = r^2 \sin^2 \theta \frac{\partial w}{\partial y} + r^2 \cos^2 \theta \frac{\partial w}{\partial y} = r^2 \frac{\partial w}{\partial y}$$

or long as $r \neq 0$ (i.e. $(x, y) \neq (0, 0)$)

we have $\rightarrow$

e.g. $\phi = \frac{kQ}{r}$ define $\vec{E} = \left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y} \right) = ?$

$$\frac{\partial \phi}{\partial r} = -\frac{kQ}{r^2}$$

$$\frac{\partial \phi}{\partial \theta} = 0$$

$$\cos \theta = \frac{x}{r} \quad \sin \theta = \frac{y}{r}$$

$$\begin{aligned} \frac{\partial w}{\partial x} &= \cos \theta \frac{\partial w}{\partial r} - \frac{\sin \theta}{r} \frac{\partial w}{\partial \theta} \\ \frac{\partial w}{\partial y} &= \sin \theta \frac{\partial w}{\partial r} + \frac{\cos \theta}{r} \frac{\partial w}{\partial \theta} \end{aligned}$$

$$\left(\left(-\frac{kQx}{(x^2+y^2)^{3/2}} \right), \left(-\frac{kQy}{(x^2+y^2)^{3/2}} \right) \right)$$

Parametric curves

def: A path (or parametrized/parametric curve) in $\mathbb{R}^n$ is a continuous function $\vec{x}: I \subseteq \mathbb{R} \rightarrow \mathbb{R}^n$.

I is an interval (e.g. $[a, b]$, $(a, b]$, $(-\infty, b)$, ...)

If $I = [a, b]$, then we call $\vec{x}(a)$ and $\vec{x}(b)$ the endpoints of the path
initial point $\rightarrow$ $\leftarrow$ terminal point

e.g.: $\vec{x}(t) = (1, 2) + t(3, -1)$ for $t \in \mathbb{R}$ (i.e. $t \in I = (-\infty, \infty)$)
is a parametrized line $\vec{v}_1 = (3, 1)$ $\vec{x}(1) = \vec{x}(0)$

$\vec{y}(t) = (4, 1) + t(-6, 2)$ for $t \in \mathbb{R}$ $-2\vec{v}_1 = \vec{v}_2 = (-6, 2)$
is another parametrization of the same line

$\vec{z}(t) = (1, 2) + \ln(t)(3, -1)$ for $t \in (0, \infty)$
is another parametrization of the same line

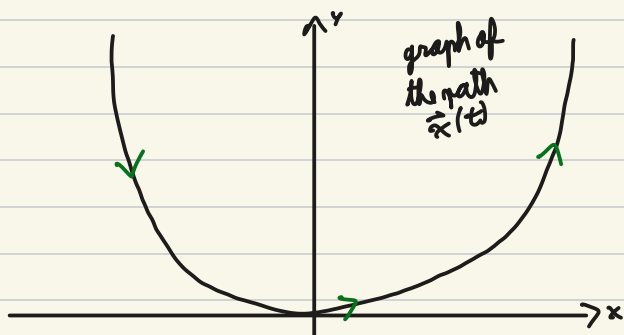
* Caution: The curve traced by $\vec{x}(t)$, $\vec{y}(t)$ and $\vec{z}(t)$ are the same, but there are all different paths (i.e. different parametrized curves, or, in other words, different parametrization of the same curve)

More examples:

e.g. $\vec{x}: [1, 5] \rightarrow \mathbb{R}^2$ given by $\vec{x}(t) = (0, 1) + t(1, -1)$
is a line segment with

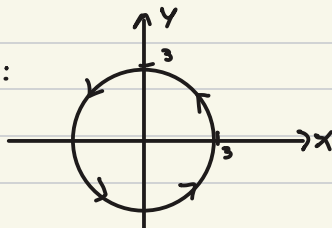
initial point $(1, 0)$ and terminal point $(5, -4)$

e.g. $\vec{x}: \mathbb{R} \rightarrow \mathbb{R}^2$ given by $\vec{x}(t) = (t, t^2)$
is a parametrization of the parabola $y = x^2$



* We can add arrows to the graph of the path following the convention that the arrow points in the direction the parameter increases

e.g. Give a parametrization of:



- the circle of radius 3 centered at (0,0) such that the path goes counterclockwise and traces the curve twice:

$$\vec{x}(t) = (3 \cos t, 3 \sin t) \quad 0 \leq t \leq 4\pi$$

endpoint: (3,0) (initial and terminal)

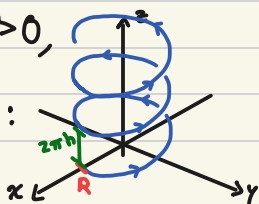
- same as above but change the endpoints

$$\vec{y}(t) = (3 \cos(t+\pi), 3 \sin(t+\pi)) \quad 0 \leq t \leq 4\pi$$

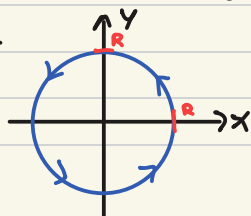
endpoint: (-3,0)

e.g. Helix $\vec{x}(t) = (R \cos t, R \sin t, ht)$ (R, h non-zero constants)

if $R > 0, h > 0$,
helix
looks like:

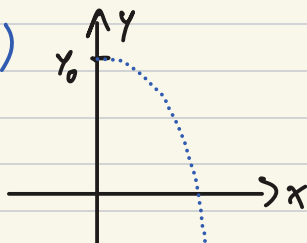


"shadow" in x-y plane
(or top view):



We typically think of the parameter t as time,
so a dynamic (changing with time) trajectory can be described by a path

e.g. $\vec{x}(t) = (t, -\frac{9.8}{2} t^2 + y_0)$



def: If $\vec{x}(t)$ is a differentiable path in $\mathbb{R}^n$
(i.e. the function $\vec{x}: I \subseteq \mathbb{R} \rightarrow \mathbb{R}^n$ is differentiable), then we
define the velocity at time t by

$$\vec{v}(t) = \vec{x}'(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{x}(t + \Delta t) - \vec{x}(t)}{\Delta t}$$

and the speed by $\|\vec{v}(t)\|$.

If $\vec{v}(t) = \vec{x}'(t)$ is differentiable (or a function $\vec{v}: I \subseteq \mathbb{R} \rightarrow \mathbb{R}^n$ $t \mapsto \vec{v}(t)$),
then we define the acceleration at time t by

$$\vec{a}(t) = \vec{v}'(t) = \vec{x}''(t)$$

If $\vec{v}(t_0) \neq \vec{0}$ (vector $(0, \dots, 0)$ in $\mathbb{R}^n$), then the tangent to the path
at $t = t_0$ is the line passing through $\vec{x}(t_0)$ in the direction of $\vec{v}(t_0)$.

e.g. $\vec{x}(t) = (4t, -5t^2 + 3t)$ at $t_0 = 1$
for $0 \leq t \leq 2$

$$\vec{v}(t) = (4, -10t + 3)$$

$$\vec{a}(t) = (0, -10)$$

$$\vec{v}(1) = (4, -7)$$

$$\vec{a}(1) = (0, -10)$$

